Lec 03

# Statistics

learning from

Academic discipline dealing with all aspects of data

quantification

## Perspective:-

- art of summarizing data
  - ↳ make data comprehensible
- science of uncertainty
  - ↳ most information in the world is uncertain
- Science of decisions
  - ↳ ultimate goal of statistics

- science of variation
  - ↳ central tendency and speed
- art of forecasting
  - ↳ future decisions on base of that data
- Science of measurement and data collection.

# Foundations of data

## Source of data

- Designed data — "artifically collected" (surveys, studies etc)
- Organic data (process generated)

For both data needs to be i.i.d
"independent", "identically distributed"
↳ more on this later!

Q: what is the source of NHANES data?
Designed

Some data is not numeric!

for instance race or gender

Just as we have data types in programming languages, we have different types here.

Weight —— numeric, continuous

# of kids —— numeric, discrete

Age group $(\overset{0}{\text{child}}, \overset{1}{\text{adult}}, \overset{2}{\text{elder}})$ —— categorical, ordered

Gender —— categorical, unordered

**imp in ML**: Practical note:-

Gender represented as: M / F

or: 0 / 1 → But still unordered

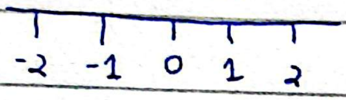
or:

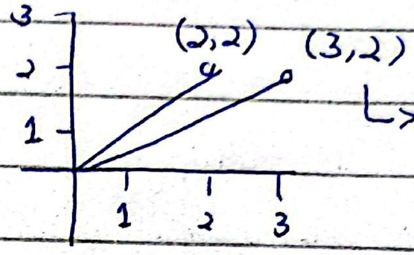
$$\begin{array}{cc} M & F \\ \begin{bmatrix} 1 \\ 0 \end{bmatrix} & \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{array}$$

$\begin{bmatrix} 1 \\ 0 \end{bmatrix} / \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ → now unordered

"One hot vector representation"

$0 < 1$




↳ they are not compareable

$(2,2) < (3,2)$

& not compareable

**Variables**

- **Quantitative** (numeric)
  - Continuous
  - Discrete
    avg make sence
- **Qualitative** (categorical)
  - Ordinal
    (ranks)
    ↳ avg no sence
  - Nominal
    (no ranks)

Discrete, Ordinal: Both discrete and ordered So, difference?